



Multivariate statistical analysis

Prof. Xuejun Jiang

- Department of Statistics and Data Science
- Office: R306, Business School Building (BS Bld)
- Office hour: Wednesday, 14:00-16:00, R306, BS Bld
- Email: jiangxj@sustech.edu.cn



Bibliography

- Textbook:
 - 1) 应用多元分析（第四版）, 王学民编著
 - 2) Applied Multivariate Statistical Analysis, Johnson RA and Wichern DW, Pearson
- Reference

Modern Multivariate Statistical Techniques, G. Casella S. Fienberg I. Olkin, Spring
- Prerequisites: Advanced mathematics, Linear algebra, Probability, mathematical Statistics

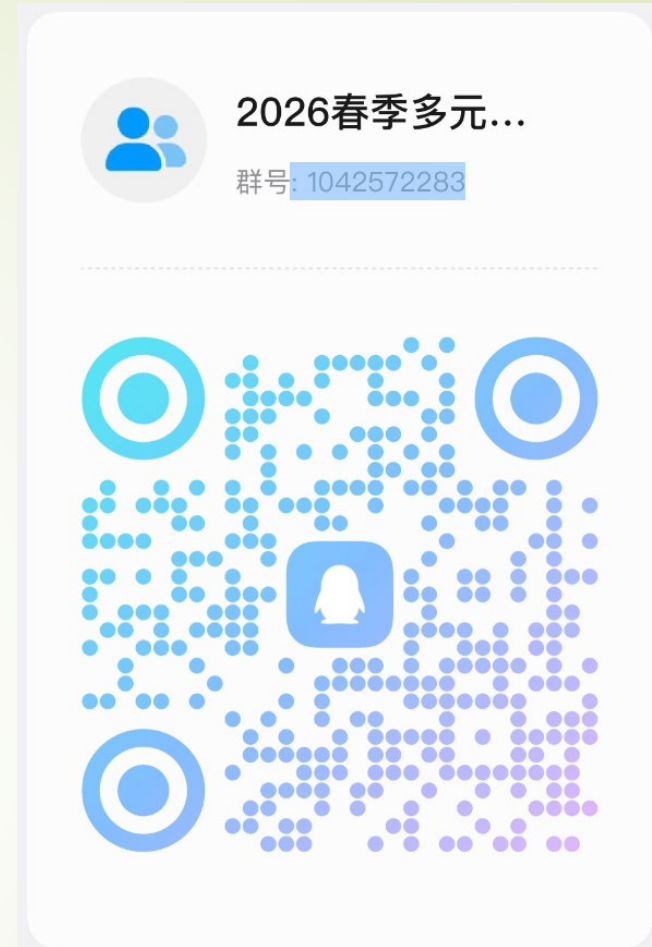


Approximate Class Agenda

- Ch1 Preliminary introduction to Matrix Analysis :
- Ch2 Statistical inference on Multivariate normal population
- Ch3 Discriminant analysis
- Ch4 Cluster analysis
- Ch5 Principal component analysis
- Ch6 Factor analysis
- Group project (Week 11 or 12)
- Ch7 Canonical correlation analysis
- Ch8 Selected topics on modern multivariate statistical methods
 - Artificial Neural Networks; Support Vector Machines
- Final exam

Course arrangement

- Grading
 - ▶ Attendance and Assignment 30%
 - ▶ Project 20%
 - ▶ Final Exam 50%
- Blackboard Course
- Course QQ group number: 1042572283
- Teaching Assistant
 - ▶ 余成耀, 17857686439 (12532239@mail.sustech.edu.cn)
 - ▶ 孙凯璐, 15028188571 (12532252@mail.sustech.edu.cn)
 - ▶ 梁荣梅, 15820760653 (12531172@mail.sustech.edu.cn)



Have Happy Cooperation!



High marks

Chapter 1 Preliminary introduction to Matrix Analysis

- §1.1 Several commonly used special matrices
- §1.4 Matrix Inverse + Generalized inverse
- §1.5 Order
- §1.6 Eigenvalue、 Eigenvector & Trace
- §1.6⁺ Singular value decomposition, generalized inverse
- §1.7 Positive Definite Matrix and Non-negative Definite Matrix
- §1.8 Extremal Problem of Eigenvalue
- §1.9 Application

§ 1.1 Several commonly used special matrices

➤ **Orthogonal:** If two p -dimensional vectors satisfy

$$\mathbf{a}'\mathbf{b}=a_1b_1+a_2b_2+\cdots+a_pb_p=0$$

then $\mathbf{a}$ and $\mathbf{b}$ are orthogonal. Geometrically, orthogonal vectors are mutually orthogonal.

➤ **Orthogonal(正交)Matrix:** if square matrix $\mathbf{A}$ satisfies $\mathbf{A}\mathbf{A}'=\mathbf{A}'\mathbf{A}=\mathbf{I}$.

- p row vectors are unit vectors and mutually orthogonal;
- p column vectors are mutually orthogonal;
- p column vectors are also a group of orthogonal unit vectors.

➤ **Idempotent (幂等) Matrix:** if square matrix $\mathbf{A}$ satisfies $\mathbf{A}^2=\mathbf{A}$.

➤ **Projection(投影) Matrix:** symmetric idempotent matrix, denoted as

$$P_M = M(M^T M)^{-1}M^T \text{ for a full column rank } l \times m \text{ matrix } M$$

$$P_M = M^T(MM^T)^{-1}M \text{ for a full row rank } m \times l \text{ matrix } M$$

► **Projection(投影) Matrix:** symmetric idempotent matrix, denoted as
(and hat matrix) $P_M = M(M^T M)^{-1} M^T$ for a full column rank
projection matrix is also name as hat matrix

Example 1 : For linear regression model
 $y = X\beta + \varepsilon$.

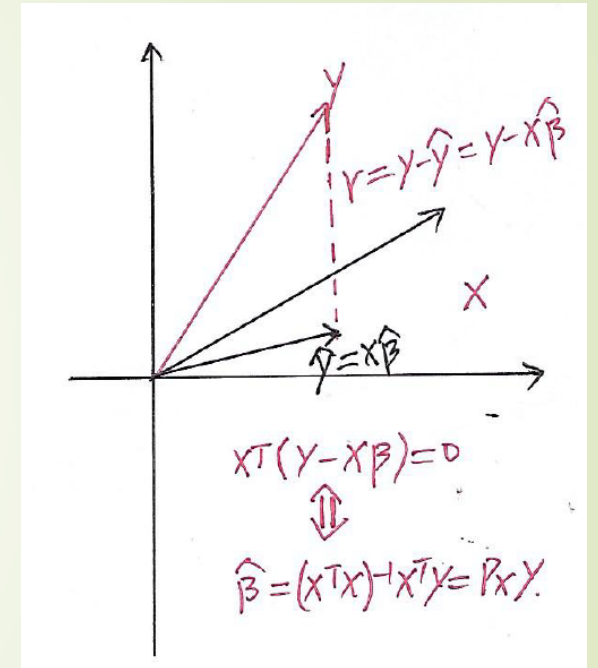
The fitted value of response vector is

$$\hat{y} = (X^T X)^{-1} X^T y = P_X y$$

in the sense of least squares. Here, $P_X y$ maps
 y onto the column space of X .

The vector of residual r can be expressed as

$$r = y - \hat{y} = y - P_X y = (I - P_X) y.$$



§ 1.4 Matrix Inverse

- **Nondegenerate Square Matrix:** for square matrix A , $|A| \neq 0$;
- **Degenerate Square Matrix:** for square matrix A , $|A| = 0$.
- **Inverse Matrix:** for a nonsingular square matrix $A = (a_{ij})$, if square matrix C satisfies $AC = I$, we can call C the inverse matrix A , $C = A^{-1}$, A^{-1} is a nondegenerate square matrix. Let

$$B = (A_{ij}) / |A|$$

where (A_{ij}) s called adjoint matrix (伴随矩阵) with A_{ij} being the algebraic complement of a_{ij} .

Basic Features of Inverse Matrix

➤ (1) $AA^{-1}=A^{-1}A=I$.

➤ (2) $(A')^{-1}=(A^{-1})'$.

➤ (3) If A and C are p dimensional nondegenerate square matrix,

$$(AC)^{-1}=C^{-1}A^{-1}$$

➤ (4) $|A^{-1}|=|A|^{-1}$.

➤ (5) If A is orthogonal, $A^{-1}=A'$.

➤ (6) If $A=\text{diag}(a_{11},a_{22},\cdots,a_{pp})$ is nondegenerate(i.s. $a_{ii}\neq 0, \quad i=1,2,\cdots,p$),

$$A^{-1} = \text{diag}(a_{11}^{-1}, a_{22}^{-1}, \cdots, a_{pp}^{-1})$$

➤ (7) If A and B are nondegenerate square matrix,

$$\begin{pmatrix} A & \mathbf{0} \\ \mathbf{0} & B \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & \mathbf{0} \\ \mathbf{0} & B^{-1} \end{pmatrix}$$

Basic Features of Inverse Matrix

➡ (8) $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$, then

$$A^{-1} = \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1} A_{12} D^{-1} A_{21} A_{11}^{-1} & -A_{11}^{-1} A_{12} D^{-1} \\ -D^{-1} A_{21} A_{11}^{-1} & D^{-1} \end{pmatrix},$$

where $D = A_{22} - A_{21} A_{11}^{-1} A_{12}$.

Proof: A^{-1} can be easily obtained from the following elementary row operations

$$\begin{pmatrix} I_r & 0 \\ -A_{21} A_{11}^{-1} & I_s \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} \\ 0 & D \end{pmatrix}$$

and

$$\begin{pmatrix} I_r & -A_{12} D^{-1} \\ 0 & I_s \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ 0 & D \end{pmatrix} = \begin{pmatrix} A_{11} & 0 \\ 0 & D \end{pmatrix}$$

Basic Features of Inverse Matrix

► Proof (continue): Combination of the above two equations implies

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}^{-1} = \begin{pmatrix} A_{11} & \\ 0 & D \end{pmatrix}^{-1} \begin{pmatrix} I_r & -A_{12}D^{-1} \\ 0 & I_s \end{pmatrix} \begin{pmatrix} I_r & 0 \\ -A_{21}A_{11}^{-1} & I_s \end{pmatrix}$$

That is

$$A^{-1} = \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}D^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}D^{-1} \\ -D^{-1}A_{21}A_{11}^{-1} & D^{-1} \end{pmatrix}.$$

§ 1.5 Order

- Given a group of same dimensional vector $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, if there exist not complete zero constants $c_1, c_2, \dots, c_n$ such that

$$c_1 \mathbf{a}_1 + c_2 \mathbf{a}_2 + \dots + c_n \mathbf{a}_n = \mathbf{0}$$

the group of vectors are **Linear Dependent**. If vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are not linear correlated, then we call them **Linear Independent**.

- **Row Rank:** the maximum number of linear independence (线性无关) row vectors of matrix;
- **Column Rank:** the maximum number of linearly independent columns of matrix.
- Row rank and column rank of a certain matrix will be equal, so we call it **Rank of A**, $\text{rank}(A)$.

Basic Features of Rank

- (1) $\text{rank}(\mathbf{A})=0$ if and only if $\mathbf{A}=\mathbf{0}$.
- (2) Assume $\mathbf{A}$ is a $p \times q$ matrix and $\mathbf{A} \neq \mathbf{0}$, $1 \leq \text{rank}(\mathbf{A}) \leq \min\{p, q\}$
(If $\text{rank}(\mathbf{A})=p$, $\mathbf{A}$ is Row Full Rank; If $\text{rank}(\mathbf{A})=q$, $\mathbf{A}$ is Column Full Rank).
- (3) $\text{rank}(\mathbf{A})=\text{rank}(\mathbf{A}')$ ($\Leftarrow$ (8)).
- (4) $\text{rank} \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{pmatrix} = \text{rank} \begin{pmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{B} & \mathbf{0} \end{pmatrix} = \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B})$
- (5) $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$.
- (6) If $\mathbf{A}$ and $\mathbf{C}$ are nondegenerate square matrix,
$$\text{rank}(\mathbf{ABC}) = \text{rank}(\mathbf{B})$$
- (7) p dimensional matrix $\mathbf{A}$ is nondegenerate, if and only if $\text{rank}(\mathbf{A})=p$
(we call $\mathbf{A}$ Full Rank).

Basic Features of Rank

➡ (8) $\text{rank}(AA') = \text{rank}(A'A) = \text{rank}(A)$ (Exercise).

Proof: Let A be a $p \times q$ matrix, then the dimensions of solution space for the system of linear equation $Ax = 0$ and $A'Ax = 0$ are

$$q - \text{rank}(A) \text{ and } q - \text{rank}(A'A),$$

respectively.

If x is the solution of $Ax = 0$, then x is also the solution of $A'Ax = 0$. Conversely, if x is the solution of $A'Ax = 0$, then $(Ax)'Ax = x'(A'Ax) = 0$ which leads to $Ax = 0$. So, the systems of $Ax = 0$ and $A'Ax = 0$ have the same solution space. This implies

$$q - \text{rank}(A) = q - \text{rank}(A'A).$$

Therefore, $\text{rank}(A'A) = \text{rank}(A)$.



§ 1.6 Eigenvalue, Eigenvector & Trace

- 1.6.1 Eigenvalue & Eigenvector
- 1.6.2 Trace

1.6.1 Eigenvalue & Eigenvector

- Assume that A is a p dimensional square matrix, if there exists a p dimensional non-zero vector $\mathbf{x}$ with respect to a certain parameter λ such that $A\mathbf{x}=\lambda\mathbf{x}$, λ is called Eigenvalue of A and $\mathbf{x}$ is called Eigenvector with respect to eigenvalue λ .
- According to the definition above, $(A-\lambda I)\mathbf{x}=\mathbf{0}$, and, $\mathbf{x}\neq\mathbf{0}$, we can obtain

$$|A-\lambda I|=0,$$

which is a p degree polynomial of λ , called **Characteristic Polynomial**. $|A-\lambda I|=0$ has p roots (some may be multiple roots), denoted by $\lambda_1, \lambda_2, \dots, \lambda_p$. Vice versa, if λ_i is one root of $|A-\lambda I|=0$, then $A-\lambda_i I$ is a degenerate matrix. There will exist a p dimensional non-zero vector $\mathbf{x}_i$ such that

$$(A-\lambda_i I)\mathbf{x}_i=\mathbf{0},$$

i.e., λ_i is an eigenvalue of A and $\mathbf{x}_i$ is the corresponding eigenvector.

- Usually, $\mathbf{x}_i$ is set as a unit vector, i.e., $\mathbf{x}_i'\mathbf{x}_i=1$.

Basic Features of Eigenvalue & Eigenvector

- (1) A and A' have the same eigenvalues.
- (2) If A and B are separately $p \times q$ and $q \times p$ matrix, then AB and BA have the same non-zero eigenvalues.

Proof: From

$$\begin{pmatrix} I_p & -A \\ 0 & \lambda I_q \end{pmatrix} \begin{pmatrix} \lambda I_p & A \\ B & I_q \end{pmatrix} = \begin{pmatrix} \lambda I_p - AB & 0 \\ \lambda B & \lambda I_q \end{pmatrix} \text{ and } \begin{pmatrix} I_p & 0 \\ -B & \lambda I_q \end{pmatrix} \begin{pmatrix} \lambda I_p & A \\ B & I_q \end{pmatrix} = \begin{pmatrix} \lambda I_p & A \\ 0 & \lambda I_q - BA \end{pmatrix},$$

we can obtain

$$\left| \begin{pmatrix} \lambda I_p - AB & 0 \\ \lambda B & \lambda I_q \end{pmatrix} \right| = \left| \begin{pmatrix} \lambda I_p & A \\ 0 & \lambda I_q - BA \end{pmatrix} \right|.$$

This leads to $\lambda^q |\lambda I_p - AB| = \lambda^p |\lambda I_q - BA|$. Therefore, two equations about λ , $|\lambda I_p - AB|$ and $|\lambda I_q - BA|$, have exactly the same nonzero eigenvalues. Furthermore, if there are nonzero eigenvalues, their multiplicities are demonstrated to be the same.

Basic Features of Eigenvalue & Eigenvector

- (3) If A is real symmetric matrix, all eigenvalues of A are real numbers, we order them from the largest to the smallest as $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$. If $\lambda_i \neq \lambda_j$, then the corresponding eigenvectors $\mathbf{x}_i$ and $\mathbf{x}_j$ will be orthogonal, as $\mathbf{x}_i' \mathbf{x}_j = 0$.

(不同特征值的特征向量彼此正交)

- (4) If $A = \text{diag}(a_{11}, a_{22}, \dots, a_{pp})$, then $a_{11}, a_{22}, \dots, a_{pp}$ are p number of eigenvalues of A , corresponding eigenvectors are separately $\mathbf{e}_1 = (1, 0, \dots, 0)'$, $\mathbf{e}_2 = (0, 1, 0, \dots, 0)'$, $\dots$, $\mathbf{e}_p = (0, \dots, 0, 1)'$.

- (5) $|A| = \prod_{i=1}^p \lambda_i$, i.e., the determinant of A equals to the product of its eigenvalues. A is nondegenerate, if and only if none of the eigenvalues are zero; A is degenerate, if and only if at least one of the eigenvalues is zero.

Example 1.6.4 (Useful conclusion). Let $\lambda_1, \lambda_2, \dots, \lambda_p$ are p eigenvalues of a $p \times p$ matrix A . Try to prove:

(i) If A is reversible, λ_i is a eigenvalue of A with corresponding eigenvectors of x_i , then $1/\lambda_i$ is the eigenvalues of A^{-1} with the same eigenvector x_i ;

Proof: for $x \neq 0$, $Ax = \lambda x \Rightarrow x = \lambda A^{-1}x \Rightarrow A^{-1}x = \lambda^{-1}x$

(ii) If A is idempotent matrix, the eigenvalues of A are 0 or 1;

Proof: for $x \neq 0$, $\lambda x = Ax = A^2x = A\lambda x = \lambda Ax = \lambda^2x$.

(iii) If A is orthogonal matrix, the eigenvalues of A are -1 or 1.

Proof: for $x \neq 0$, $x = A'Ax = A'\lambda x = \lambda A'x = \lambda^2x$.

(6) If A is p dimensional symmetric matrix, then there exist orthogonal matrix T and diagonal matrix $\Lambda = (\lambda_1, \lambda_2, \dots, \lambda_p)$ such that

$$A = T\Lambda T' \rightarrow \text{实对称矩阵合同于对角矩阵}$$

Multiply T to the above equation leads to

$$AT = T\Lambda$$

Denote by $T = (t_1, t_2, \dots, t_p)$, we have

$$(At_1, At_2, \dots, At_p) = (\lambda_1 t_1, \lambda_2 t_2, \dots, \lambda_p t_p).$$

So $At_i = \lambda_i t_i$, $i=1, 2, \dots, p$. This suggests that $\lambda_1, \lambda_2, \dots, \lambda_p$ is eigenvalues of A and $t_1, t_2, \dots, t_p$ are corresponding eigenvector (a group of orthogonal unit vector).

Spectral Factorization (谱分解) Theorem:

$$A = T\Lambda T' = \sum_{i=1}^p \lambda_i t_i t_i'.$$

➤ If A is a square matrix with full rank, **Spectral Factorization can be used to find the inverse of A or the rank of A .**

Diagonalization of the real symmetrical matrix

Procedures:

1. Finding the eigenvalues of A : $\lambda_1, \lambda_2, \dots, \lambda_p$ by solve

$$|\lambda I - A| = 0.$$

2. Solve the basic solution system of $(\lambda_i I - A)x = 0$;
3. Apply the **Gram-Schmidt orthogonalization process** to the basic solution system to obtain p linearly independent unit eigenvectors $\gamma_1, \gamma_2, \dots, \gamma_p$;
4. Let $C = (\gamma_1, \gamma_2, \dots, \gamma_p)$. Then, $C^T A C = \Lambda$.

Example 1. For the symmetric matrix

$$A = \begin{bmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{bmatrix}.$$

Find an orthogonal matrix C such that $C^{-1}AC$ is a diagonal matrix.

$$\begin{aligned} \text{Solution: Solve } |\lambda I - A| &= \begin{vmatrix} \lambda - 2 & 2 & 0 \\ 2 & \lambda - 1 & 2 \\ 0 & 2 & \lambda \end{vmatrix} \\ &= (\lambda - 4)(\lambda - 2)(\lambda - 1) \\ &= 0, \end{aligned}$$

we obtain $\lambda_1 = 4, \lambda_2 = 1, \lambda_3 = -2$.

Next, we solve the basic solution system of $(\lambda_i I - A)x = 0$ for $\lambda_1 = 4, \lambda_2 = 1, \lambda_3 = -2$.

For $\lambda_1 = 4$, by $(4I - A)X = 0$, we have

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & 2 & 0 \\ 2 & \lambda - 1 & 2 \\ 0 & 2 & \lambda \end{bmatrix}$$

$$4I - A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 3 & 2 \\ 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 = 2x_3 \\ x_2 = -2x_3 \end{cases}$$

We obtain the basic solution system $\alpha_1 = \begin{bmatrix} -2 \\ 2 \\ -1 \end{bmatrix}$.

For $\lambda_2 = 1$, by $(I - A)X = 0$, we have $\lambda I - A = \begin{bmatrix} \lambda - 2 & 2 & 0 \\ 2 & \lambda - 1 & 2 \\ 0 & 2 & \lambda \end{bmatrix}$

$$I - A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix} \Rightarrow \begin{cases} x_1 = -x_3 \\ x_2 = -\frac{1}{2}x_3 \end{cases}$$

Thus, We obtain the basic solution system $\alpha_2 = (2 \ 1 \ -2)'$.

For $\lambda_3 = -2$, by $-2I - A = 0$, we have

$$-2I - A = \begin{bmatrix} -4 & 2 & 0 \\ 2 & -3 & 2 \\ 0 & 2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 = \frac{1}{2}x_3 \\ x_2 = x_3 \end{cases}$$

Thus, we obtain the basic solution system

$$\alpha_3 = (1 \ 2 \ 2)'$$

Orthogonalize the eigenvectors $\alpha_1 = \begin{bmatrix} -2 \\ 2 \\ -1 \end{bmatrix}$, $\alpha_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$, $\alpha_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

Since $\alpha_1, \alpha_2, \alpha_3$ are the eigenvectors of A corresponding to three different eigenvalues, they are orthogonal to each other. We utilize them by setting $\gamma_i = \alpha_i / \|\alpha_i\|$ for $i = 1, 2, 3$, which are listed as follows.

$$\gamma_1 = \begin{bmatrix} -2/3 \\ 2/3 \\ -1/3 \end{bmatrix}, \quad \gamma_2 = \begin{bmatrix} 2/3 \\ 1/3 \\ -2/3 \end{bmatrix}, \quad \gamma_3 = \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}.$$

Let $C = (\gamma_1, \gamma_2, \gamma_3)$. Then, we have

$$C^{-1}AC = C'AC = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

Diagonalization of the real symmetrical matrix

Gram-Schmidt orthogonalization process:

1. Suppose that $\alpha_1, \alpha_2, \alpha_n$ be a linearly independent vector group in Euclidean space. Let

$$\beta_1 = \alpha_1, \beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1, \dots, \beta_s = \alpha_s - \sum_{i=1}^{s-1} \frac{(\alpha_s, \beta_i)}{(\beta_i, \beta_i)} \beta_i$$

for $s = 2, \dots, n$.

2. Then, $\beta_1, \beta_2, \dots, \beta_s$ is a group of orthogonal unit vectors such that $\langle \alpha_1, \alpha_2, \dots, \alpha_s \rangle = \langle \beta_1, \beta_2, \dots, \beta_s \rangle$.

➤ **Exercise 1.** Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

and write the spectral factorization.

Solution: $\lambda_1 = 4, \lambda_2 = \lambda_3 = 1$. The corresponding eigenvectors are

$$t_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, t_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, t_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}. \text{ The spectral factorization is}$$

with $T = (t_1, t_2, t_3)$ and $\Lambda = \text{diag}\{4, 1, 1\}$.

$$A = T\Lambda T' = \sum_{i=1}^3 \lambda_i \mathbf{t}_i \mathbf{t}_i'$$

1.6.2 Trace

- Assume A is p dimensional square matrix, then the sum of its diagonal elements is call the **trace** of A , denoted as $\text{tr}(A)$,

$$\text{tr}(A)=a_{11}+a_{22}+\cdots+a_{pp}$$

- Basic Features of Trace :

➤ (1) $\text{tr}(AB)=\text{tr}(BA)$. Especially, $\text{tr}(ab')=b'a$.

➤ (2) $\text{tr}(A)=\text{tr}(A')$.

➤ (3) $\text{tr}(A+B)=\text{tr}(A)+\text{tr}(B)$.

➤ (4) $\text{tr}\left(\sum_{i=1}^k A_i\right)=\sum_{i=1}^k \text{tr}(A_i)$

- (5) Let $A=(a_{ij})$ be $p \times q$ matrix, then

$$\text{tr}(A'A) = \text{tr}(AA') = \sum_{i=1}^p \sum_{j=1}^q a_{ij}^2$$

- (6) Let $\lambda_1, \lambda_2, \dots, \lambda_p$ be the eigenvalues of square matrix A , then

$$\text{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_p$$

Proof: From spectral factorization theorem, there exists orthogonal matrix T such that $A = T\Lambda T'$. Thus, $\text{tr}(A) = \text{tr}(T\Lambda T') = \text{tr}(T'T\Lambda) = \sum_{i=1}^p \lambda_i$.

- (7) If A is projection matrix (**Symmetric Idempotent Matrix**), then

$$\text{tr}(A) = \text{rank}(A).$$

Proof: $\text{tr}(A) = \sum_{i=1}^p \lambda_i$ = the number of nonzer of λ_i among all λ_i for $i = 1, \dots, p$ since the eigenvalues of A are 0 or 1, which is also the $\text{rank}(A)$.

1.6.3 Singular Value Decomposition

- **Singular Value Decomposition (SVD) Theorem:** If A is a $m \times n$ matrix with $\text{rank}(A) = r$, there exist orthogonal $m \times m$ and $n \times n$ matrices P and Q exist, such that

$$A = PDQ' \text{ and } D = P'AQ,$$

$$P'AQ = \begin{bmatrix} P'_1AQ_1 & P'_1AQ_2 \\ P'_2AQ_1 & P'_2AQ_2 \end{bmatrix} \quad (6.1a)$$

where $m \times n$ matrix D is given by

(a) Λ if $r = m = n$,

(b) $\begin{bmatrix} \Lambda & 0_{r \times (n-r)} \end{bmatrix}$ if $r = m < n$,

(c) $\begin{bmatrix} \Lambda \\ 0_{(m-r) \times r} \end{bmatrix}$ if $r = n < m$

(d) $\begin{bmatrix} \Lambda & 0_{r \times (n-r)} \\ 0_{(m-r) \times r} & 0_{(m-r) \times (n-r)} \end{bmatrix}$ if $r < m, r < n$

and $\Lambda = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_r\}$. Here $\lambda_1^2, \lambda_2^2, \dots, \lambda_r^2$ are the r positive eigenvalues of $A'A$ and AA' , and $\lambda_1, \lambda_2, \dots, \lambda_r$ are called as singular values of A .

➤ $A'A = QD^2Q'$ and $AA' = PD^2P'$;

➤ If we partition $P = [P_1, P_2]$ and $Q = [Q_1, Q_2]$, then $P'_2AQ_1 = 0$ and $P'_1AQ_2 = 0$.

1.6.3 Singular Value Decomposition

- If we again partition P and Q as $P = [P_1, P_2]$ and $Q = [Q_1, Q_2]$, where P_1 is the $m \times r$ matrix and Q_1 is the $n \times r$ matrix, then the singular value decomposition can be restated as follows.

$$A = [P_1, P_2] \begin{bmatrix} \Lambda & 0_{r \times (n-r)} \\ 0_{(m-r) \times r} & 0_{(m-r) \times (n-r)} \end{bmatrix} \begin{bmatrix} Q_1' \\ Q_2' \end{bmatrix} = P_1 \Lambda Q_1'. \quad (6.1b)$$

- Denote $P_1 = [u_1, u_2, \dots, u_r]$ and $Q_1 = [v_1, v_2, \dots, v_r]$. From equation (6.1b), we have

$$AA' = P_1 \Lambda^2 P_1' \text{ and } A'A = Q_1 \Lambda^2 Q_1'.$$

This implies P_1 and Q_1 can be obtained from the spectral decomposition of AA' and $A'A$, respectively.

- P_2 can be any matrix satisfying $P_2' P_1 = P_2' A Q_1 \Lambda^{-1} = 0, P_2' P_2 = I_{m-r}$.

What is the relationship of SVD and Spectral Factorization?

1.6.3 Singular Value Decomposition

- **Another representation of SVD:** If A is a $m \times n$ matrix of rank $r > 0$, then there exists $m \times r$ and $n \times r$ matrices P_1 and Q_1 , such that $P_1'P_1 = Q_1'Q_1 = I_r$ and

$$A = P_1 \Lambda Q_1', \quad (6.2)$$

where $\Lambda = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_r\}$ with $\lambda_1^2, \lambda_2^2, \dots, \lambda_r^2$ being the r positive eigenvalues of $A'A$ and AA' .

- P_1 and Q_1 can be obtained from the spectral decomposition of AA' and $A'A$.
- SVD can be applied in matrix compression and reconstruction
- SVD can be used to find the Moore- Penrose (M-P) inverse (伪逆).

1.6.3 Singular Value Decomposition

➡ **Example 2.** Find the singular value decomposition for the 4×3 matrix

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 3 & -1 & 1 \\ -2 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Solution: First, an eigenanalysis of the matrix

$$A'A = \begin{bmatrix} 18 & -10 & 4 \\ -10 & 18 & 4 \\ 4 & 4 & 4 \end{bmatrix}$$

reveals that it has eigenvalues 28, 12, and 0 with associated normalized eigenvectors $(1/\sqrt{2}, -1/\sqrt{2}, 0)'$, $(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})'$, and $(1/\sqrt{6}, 1/\sqrt{6}, -2/\sqrt{6})'$, respectively. Let these eigenvectors be the columns of the 3×3 orthogonal matrix Q .

1.6.3 Singular Value Decomposition

► Example 2.

Solution: (continue) Clearly, $\text{rank}(A) = 2$ and the two singular values of A are $\sqrt{28}$ and $\sqrt{12}$. From $A = P_1 \Lambda Q_1'$, the 4×2 matrix P_1 is given by

$$P_1 = A Q_1 \Lambda^{-1} = \begin{bmatrix} 2 & 0 & 1 \\ 3 & -1 & 1 \\ -2 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{3} \\ 0 & 1/\sqrt{3} \end{bmatrix} \times \begin{bmatrix} 1/\sqrt{28} & 0 \\ 0 & 1/\sqrt{12} \end{bmatrix}$$
$$= \begin{bmatrix} 1/\sqrt{14} & 1/2 \\ 2/\sqrt{14} & 1/2 \\ -3/\sqrt{14} & 1/2 \\ 0 & 1/2 \end{bmatrix}$$

Q_1 can be obtained from the **spectral factorization** of $A' A$

The 4×2 matrix P_2 can be any matrix satisfying $P_1' P_2 = (0)$ and $P_2' P_2 = I_2$; for instance, $(1/\sqrt{12}, 1/\sqrt{12}, 1/\sqrt{12}, -3/\sqrt{12})'$ and $(-5/\sqrt{42}, 4/\sqrt{42}, 1/\sqrt{42}, 0)'$

can be chosen as P_2 .

1.6.3 Singular Value Decomposition

- Solution: (continue) Thus, the singular value decomposition of A in (6.1a) is given by

$$\begin{aligned}
 P \leftarrow & \begin{bmatrix} 1/\sqrt{14} & 1/2 & 1/\sqrt{12} & -5/\sqrt{42} \\ 2/\sqrt{14} & 1/2 & 1/\sqrt{12} & 4/\sqrt{42} \\ -3/\sqrt{14} & 1/2 & 1/\sqrt{12} & 1/\sqrt{42} \\ 0 & 1/2 & -3/\sqrt{12} & 0 \end{bmatrix} \begin{bmatrix} \sqrt{28} & 0 & 0 \\ 0 & \sqrt{12} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow D \\
 & \times \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \end{bmatrix}, \rightarrow Q'
 \end{aligned}$$

or in the form of (6.1b)

$$\begin{aligned}
 P_1 \leftarrow & \begin{bmatrix} 1/\sqrt{14} & 1/2 \\ 2/\sqrt{14} & 1/2 \\ -3/\sqrt{14} & 1/2 \\ 0 & 1/2 \end{bmatrix} \underbrace{\begin{bmatrix} \sqrt{28} & 0 \\ 0 & \sqrt{12} \end{bmatrix}}_{\Lambda} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \rightarrow Q'_1
 \end{aligned}$$

1.6.3 Singular Value Decomposition

Implementation of SVD:

matlab codes:

```
A=[1, 2; 3, 4; 5, 6];
```

```
[U, S, V] = svd(A);
```

```
disp('U:');
```

```
disp(U);
```

```
disp('S:');
```

```
disp(diag(diag(S))); % employ diag(diag(S)) to retain only its diagonal elements of S.
```

```
disp('V':');
```

```
disp(V');
```

```
U:
    -0.2298    0.8835    0.4082
    -0.5247    0.2408   -0.8165
    -0.8196   -0.4019    0.4082

S:
    9.5255         0
         0    0.5143

V':
    -0.6196   -0.7849
    -0.7849    0.6196
```

1.6.3 Singular Value Decomposition

➡ **Example 3:** Using SVD for matrix compression and reconstruction (matlab codes)

```
A = [1 2; 3 4; 5 6];
```

```
[U, S, V] = svd(A); % Select the first k singular values and their corresponding columns of U and V  
to reconstruct the matrix
```

```
k = 2;
```

```
S_k = S(1:k, 1:k); % Take the first k singular values to form a new singular value matrix
```

```
U_k = U(:, 1:k); % Take the first k columns of U
```

```
V_k = V(:, 1:k); % Take the first k columns of V, note that V is transposed, so taking a column  
actually corresponds to taking a row.
```

```
A_approx = U_k * S_k * V_k'; % Reconstruct matrix
```

```
% Display the original matrix and reconstructed matrix
```

```
disp('original matrixA:');
```

```
disp(A);
```

```
disp('reconstructed matrixA_approx:');
```

```
disp(A_approx);
```

```
>> mreconstr_svd
```

```
original matrixA:
```

1	2
3	4
5	6

```
reconstructed matrixA_approx:
```

1.0000	2.0000
3.0000	4.0000
5.0000	6.0000

1.6.3 Singular Value Decomposition

Implementation of SVD:

python codes:

```
import numpy as np
# Create an example matrix
A = np.array([[1, 2], [3, 4], [5, 6]])
# Using numpy's linalg.svd function for SVD
U, s, VT = np.linalg.svd(A)
# s is a vector that contains singular values;
# U and VT are orthogonal matrices
print("U:\n", U)
print("s:\n", s)
print("VT:\n", VT)
```

```
U:
[[-0.2298477  0.88346102  0.40824829]
 [-0.52474482  0.24078249 -0.81649658]
 [-0.81964194 -0.40189603  0.40824829]]
s:
[9.52551809 0.51430058]
VT:
[[-0.61962948 -0.78489445]
 [-0.78489445  0.61962948]]
```

1.6.3 Singular Value Decomposition

Example 4. Application examples of SVD in signal processing (python codes):

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib # can be omitted
matplotlib.use('Qt5Agg') # can be omitted

# Generate a clean signal (sine wave)
t = np.linspace(0, 1, 500)
clean_signal = np.sin(2 * np.pi * 5 * t) # 5 Hz sine wave

# Add random noise to the signal
noise = np.random.normal(0, 0.5, clean_signal.shape)
noisy_signal = clean_signal + noise

# Reshape the signal into a matrix for SVD
signal_matrix = noisy_signal.reshape(50, 10) # Reshape into a 50x10 matrix

# Perform SVD
U, S, Vt = np.linalg.svd(signal_matrix, full_matrices=False)

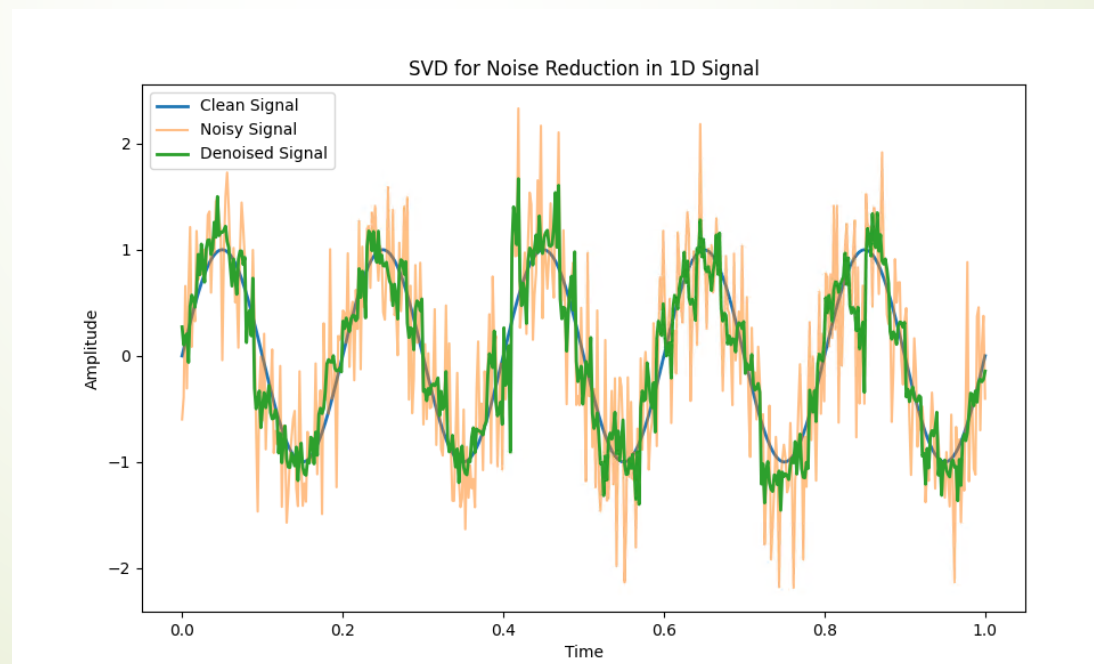
# Truncate the singular values to remove noise
k = 2 # Keep only the first k singular values
S[k:] = 0 # Set the smaller singular values to zero

# Reconstruct the denoised signal
denoised_matrix = U @ np.diag(S) @ Vt
denoised_signal = denoised_matrix.reshape(-1) # Flatten back to 1D
```

1.6.3 Singular Value Decomposition

Example 4. Application examples of SVD in signal processing (python codes):

```
# Plot the results
plt.figure(figsize=(10, 6))
plt.plot(t, clean_signal, label="Clean Signal", linewidth=2)
plt.plot(t, noisy_signal, label="Noisy Signal", alpha=0.5)
plt.plot(t, denoised_signal, label="Denoised Signal", linewidth=2)
plt.legend() # a function in the Matplotlib library used to add legends (图例)
plt.title("SVD for Noise Reduction in 1D Signal")
plt.xlabel("Time")
plt.ylabel("Amplitude")
plt.show()
```



Generalized Inverse of Matrix

- Why is it necessary to introduce generalized inverse matrices

Let us see a system of linear equations

$$Ax = b,$$

where A is a $m \times n$ matrix of constants, b is a $m \times 1$ vector of constants. How to find the solution of the above system.

- If $m = n$ and A is nonsingular, then A^{-1} exists and the solution $x = A^{-1}b$.
- However, when A^{-1} does not exist, how to determine the system has any solutions?

A is a full row rank matrix,

A is a full column rank matrix,

- The introduction to generalized inverse of matrix can help us to answer the above question.

In your linear model class, the generalized inverse A^- has been introduced to satisfy $AA^-A = A$ but A^- is not unique for this definition. Let's move on to **the Moore-Penrose (M-P) inverse (伪逆)**.

Generalized Inverse of Matrix

➤ **The Moore-Penrose (M-P) inverse**, 伪逆 A^+ , of an m -by- n matrix A is defined by the **unique** n -by- m matrix satisfying the following four criteria

1) $AA^+A = A$

2) $A^+AA^+ = A^+$

3) $(AA^+)' = AA^+$

4) $(A^+A)' = A^+A$

➤ From SVD Theorem, we know $m \times r$ matrix $n \times r$ matrices P and Q exist, such that $P_1'P_1 = Q_1'Q_1 = I_r$ and

$$A = P_1 \Lambda Q_1'.$$

Exercise 2. If we define $A^+ = Q_1 \Lambda^{-1} P_1'$, we can confirm A^+ satisfies the above four conditions. Thus, A^+ is a M-P inverse of A . Furthermore, we can prove that the above A^+ is **unique**. [Hints: suppose B and C are any two matrices satisfying condition 1-4 for A^+ . Then using these four conditions to prove that $B = C$].

Generalized Inverse of Matrix

► Exercise 2. Proof:

$$A^+ = Q_1 \Lambda^{-1} P_1'$$

Thus, $A^+ = Q\Delta^{-1}P'$ is a Moore–Penrose inverse of A , and so we have established the existence of the Moore–Penrose inverse. Next, suppose that B and C are any two matrices satisfying conditions (5.1)–(5.4) for A^+ . Then using these four conditions, we find that

$$\begin{aligned} AB &= (AB)' = B'A' = B'(ACA)' = B'A'(AC)' \\ &= (AB)'AC = ABAC = AC \end{aligned}$$

and

$$\begin{aligned} BA &= (BA)' = A'B' = (ACA)'B' = (CA)'A'B' \\ &= CA(BA)' = CABA = CA. \end{aligned}$$

Now using these two identities and (5.2), we see that

$$B = BAB = BAC = CAC = C.$$

Since B and C are identical, the Moore–Penrose inverse is unique.



Generalized Inverse of Matrix

➤ **The Moore-Penrose (M-P) inverse**, 广义(伪)逆 A^+ , of an m -by- n matrix A is defined by the **unique** n -by- m matrix satisfying the following four criteria

1) $AA^+A = A$

2) $A^+AA^+ = A^+$

3) $(AA^+)' = AA^+$

4) $(A^+A)' = A^+A$

- If A is a full row rank ($= m$), then $A^+ = A'(AA')^{-1}$, and one solution of $Ax = b$ is $x = A^+b$.
- If A is a full column rank ($= n$), then $A^+ = (A'A)^{-1}A'$, and the approximate solution of $Ax = b$ is $x = A^+b$ by noting

$$Ax = b \Rightarrow A^+Ax = A^+b \Rightarrow x = A^+b = (A'A)^{-1}A'b.$$

In this case, the solution is not exact, which is an approximate solution of $Ax = b$ in the least squares sense:

$$\min_x (Ax - b)'(Ax - b)$$

Generalized Inverse of Matrix

➤ **The Moore-Penrose (M-P) inverse**, 广义(伪)逆 A^+ , of an m -by- n matrix A is defined by the **unique** n -by- m matrix satisfying the following four criteria

1) $AA^+A = A$

2) $A^+AA^+ = A^+$

3) $(AA^+)' = AA^+$

4) $(A^+A)' = A^+A$

➤ Properties of M-P inverse:

(a) $(\alpha A)^+ = \alpha^{-1}A^+$, if $\alpha \neq 0$ is a scalar,

(b) $(A')^+ = (A^+)',$

(c) $(A^+)^+ = A,$

(d) $A^+ = A^{-1}$, if A is square and nonsingular,

(e) $(A'A)^+ = A^+A^{+'}$ and $(AA')^+ = A^{+'}A^+,$

(f) $(AA^+)^+ = AA^+$ and $(A^+A)^+ = A^+A,$

(g) $A^+ = (A'A)^+A' = A'(AA')^+,$ To confirm A^+ satisfy the above four criteria.

(h) $A^+ = (A'A)^{-1}A'$ and $A^+A = I_n$, if $\text{rank}(A) = n,$

(i) $A^+ = A'(AA')^{-1}$ and $AA^+ = I_m$, if $\text{rank}(A) = m,$

(j) $A^+ = A'$, if the columns of A are orthogonal, that is, $A'A = I_n.$

Generalized Inverse of Matrix

Example 5. Properties (h) and (i) of M-P inverse give useful ways of computing the M-P inverse of matrices that have full column rank or full row rank. We demonstrate this by finding the M-P inverse of

$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix}.$$

From property (h), for any vector $a \neq 0$, a^+ will be given by $(a'a)^{-1}a'$, so here we find that $a^+ = [0.5 \ 0.5]$.

For A , we can use property (i) because $\text{rank}(A) = 2$. Computing AA' and $(AA')^{-1}$, we get

$$AA' = \begin{bmatrix} 6 & 4 \\ 4 & 5 \end{bmatrix}, \quad (AA')^{-1} = \frac{1}{14} \begin{bmatrix} 5 & -4 \\ -4 & 6 \end{bmatrix},$$

and so

$$\begin{aligned} A^+ &= A'(AA')^{-1} = \frac{1}{14} \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 5 & -4 \\ -4 & 6 \end{bmatrix} \\ &= \frac{1}{14} \begin{bmatrix} -3 & 8 \\ 6 & -2 \\ 5 & -4 \end{bmatrix}. \end{aligned}$$

Generalized Inverse of Matrix

► **Example 6.** To find a M-P inverse of the following symmetric matrix

$$A = \begin{bmatrix} 32 & 16 & 16 \\ 16 & 14 & 2 \\ 16 & 2 & 14 \end{bmatrix}.$$

Solution: It is easy to obtain two positive eigenvalues 48 and 12. Using SVD theorem ($A = P_1 \Lambda Q_1'$ similar to Example 2), we can obtain

$$A = \begin{bmatrix} 2/\sqrt{6} & 0 \\ 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 48 & 0 \\ 0 & 12 \end{bmatrix} \begin{bmatrix} 2/\sqrt{6} & 1/\sqrt{6} & 1/\sqrt{6} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

Thus, $A^+ (= Q_1 \Lambda^{-1} P_1')$ is defined as

$$\begin{aligned} A^+ &= \begin{bmatrix} 2/\sqrt{6} & 0 \\ 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/48 & 0 \\ 0 & 1/12 \end{bmatrix} \begin{bmatrix} 2/\sqrt{6} & 1/\sqrt{6} & 1/\sqrt{6} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \\ &= \frac{1}{288} \begin{bmatrix} 4 & 2 & 2 \\ 2 & 13 & -11 \\ 2 & -11 & 13 \end{bmatrix}. \end{aligned}$$

Generalized Inverse of Matrix

► Computation for Moore-Penrose (M-P) inverse:

- In MATLAB, the `pinv()` function is used to calculate the pseudo inverse of a matrix.

```
A = [1, 2; 2, 4; 3, 6];
```

```
A_pinv = pinv(A);
```

A_pinv =

0.0143	0.0286	0.0429
0.0286	0.0571	0.0857

- **Tolerance (容差) parameter:** When calling the `pinv()` function, you can specify a tolerance value that determines which smaller singular values will be ignored, thereby affecting the calculation of the pseudo inverse. By default, MATLAB uses a tolerance value based on machine accuracy.

```
tol = 1e-10; % Set tolerance value
```

```
A_pinv = pinv(A, tol);
```

- In python: the `pinv` function in the NumPy library is used to calculate the pseudo inverse.

```
import numpy as np
```

```
A = np.array([[1, 2], [2, 4] [3, 6]])
```

```
A_pinv = np.linalg.pinv(A) % first performs SVD on A, and then  
calculates the pseudo.
```

A_pinv:

[[0.01428571	0.02857143	0.04285714]
[0.02857143	0.05714286	0.08571429]]



Exercise 3: Use the SVD formula (6.2) of the matrix

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}$$

$$A^+ = Q_1 \Lambda^{-1} P_1' \quad (6.2)$$

to find its Moore-Penrose inverse .



Application of generalized inverse in systems of linear equation

- One of the applications of generalized inverses is in finding solutions to a system of linear equations of the form

$$Ax = c \quad (6.3)$$

where A is an $m \times n$ matrix of constants, c is an $m \times 1$ vector of constants, and x is an $n \times 1$ vector of variables.

- The system of equations in (7.3) is said to be consistent (相容的) if one or more solutions exist; otherwise, the system is referred to as an inconsistent system.
 - Theorem 6.1.** The system of equation, $Ax = c$, is consistent iff $rank([A \ c]) = rank(A)$.
The following Theorem gives an alternative necessary and sufficient condition for consistency applying a generalized inverse, A^- , of A .
 - Theorem 6.2.** The system of equation, $Ax = c$, is consistent iff for some generalized, A^- , of A , $AA^-c = c$.
 - For full row rank matrix A , the above theorem holds naturally since $A^+ = A' (A A')^{-1}$.

Application of generalized inverse in systems of linear equation

- **Theorem 6.2.** The system of equation, $Ax = c$, is consistent iff for some generalized, A^- , of A , $AA^-c = c$.
 - For full row rank matrix A , the above theorem holds naturally since $A^+ = A'(AA')^{-1}$.

Proof. First, suppose that the system is consistent and x_* is a solution, so that $c = Ax_*$. Premultiplying this identity by AA^- , where A^- is any generalized inverse of A , yields

$$AA^-c = AA^-Ax_* = Ax_* = c,$$

as is required. Conversely, now suppose that there is a generalized inverse of A satisfying $AA^-c = c$. Define $x_* = A^-c$, and note that

$$Ax_* = AA^-c = c.$$

Thus, because $x_* = A^-c$ is a solution, the system is consistent, and so the proof is complete. $\square$

*Application of generalized inverse in systems of linear equation

- **Theorem 6.3.** Suppose that $Ax = c$ is a consistent system of equations, and let A^- be any generalized inverse of the $m \times n$ matrix A . Then, for any $n \times 1$ vector y ,

$$x_y = A^-c + (I_n - A^-A)y \quad (6.4)$$

is a solution, and for any solution, x_* , a vector y exists, such that $x_* = x_y$.

Proof. Since $Ax = c$ is a consistent system of equations, we know from Theorem 6.2 that $AA^-c = c$, and so

$$\begin{aligned} Ax_y &= AA^-c + A(I_n - A^-A)y \\ &= c + (A - AA^-A)y = c, \end{aligned}$$

because $AA^-A = A$. Thus, x_y is a solution regardless of the choice of y . On the other hand, if x_* is an arbitrary solution, so that $Ax_* = c$, it follows that $A^-Ax_* = A^-c$. Consequently,

$$A^-c + (I_n - A^-A)x_* = A^-c + x_* - A^-Ax_* = x_*,$$

so that $x_* = x_{x_*}$. This completes the proof. $\square$

➤ It is obvious that A^-c is a solution for the consistent system $Ax = c$.

§ 1.7 Positive Definite Matrix and Non-negative Definite Matrix

- Let A be p dimensional **symmetric matrix** and x is a p dimensional vector, then $x'Ax$ is call the Quadratic Form of A .
- If $x'Ax > 0$ about all $x \neq 0$, we call A is **Positive Definite Matrix**, denoted as $A > 0$;
- If $x'Ax \geq 0$ about all $x \neq 0$, we call A is **Non-negative Definite Matrix**, denoted as $A \geq 0$
- For non-negative definite matrix A and B , $A > B$ means $A - B > 0$, $A \geq B$ means $A - B \geq 0$.

Basic Features of Positive Definite Matrix and Non-negative Definite Matrix

- ➡ (1) Assume A is a symmetric matrix, then A is positive (non-negative) definite matrix, if and only if all eigenvalues of A are positive (non-negative).

Proof: [necessity] There exists an orthogonal matrix T such that

$$T'AT = \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p).$$

Let $e_i = (0, \dots, 0, 1, 0, \dots, 0)'$ with 1 being the i -th position in this vector. Then, we have $\lambda_i = e_i' \Lambda e_i = (Te_i)' A (Te_i)$. Since $Te_i \neq 0$, it follows that $\lambda_i > 0$ when $A > 0$ or $\lambda_i \geq 0$ when $A \geq 0$.

[sufficiency] Let $y = Tx$ for an $x \neq 0$. Then, we have

$$x'Ax = x'T'\Lambda T'x = y'\Lambda y = \sum_{i=1}^p \lambda_i y_i^2.$$

Since $y \neq 0$, it follows that $x'Ax > (\geq) 0$ when $\lambda_i > (\geq) 0$.

Basic Features of Positive Definite Matrix and Non-negative Definite Matrix

- (1) Assume A is a **symmetric matrix**, then A is positive (non-negative) definite matrix, if and only if all eigenvalues of A are positive (non-negative).
- (2) Let $A \geq 0$, then the rank of A equals to the number of positive eigenvalues of A .
- (3) If $A > 0$, then $A^{-1} > 0$.

Proof: The eigenvalues of A and A^{-1} are reciprocal to each other. From (1), it follows that (3) holds

- (4) Let $A \geq 0$, then $A > 0$, if and only if $|A| \neq 0$.

Basic Features of Positive Definite Matrix and Non-negative Definite Matrix

- (5) If $A > 0$ (or ≥ 0), then $|A| > 0$ (or ≥ 0).
- (6) $BB' \geq 0$ for all the matrix B .
- (7) If $A > 0$ (or ≥ 0), then there exists $A^{1/2} > 0$ (or ≥ 0) such that $A = A^{1/2} A^{1/2}$, $A^{1/2}$ is called the square root matrix of A .
- (8) Let $A \geq 0$ is p dimensional matrix with rank r , then there exists a $p \times r$ matrix B with rank r (column full rank) such that $A = BB'$.

Proof: It follows that by spectral factorization (谱分解) theorem.

Basic Features of Positive Definite Matrix and Non-negative Definite Matrix

- ➡ (8) Let $A \geq 0$ is p dimensional matrix with rank r , then there exists a $p \times r$ matrix B with rank r (column full rank) such that $A = BB'$.

Proof: It follows that by spectral factorization (谱分解) theorem there exists an orthogonal matrix T and diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$ to satisfy $A = T\Lambda T'$, where $\lambda_1 \geq \lambda_2 \geq \dots \lambda_{r+1} = \dots = \lambda_p = 0$. Let $\Lambda_1 = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_r)$, $T = (T_1, T_2)$ with T_1 being $p \times r$ matrix and T_2 being $p \times (p - r)$ matrix. Then, we have

$$A = T\Lambda T' = (T_1 \ T_2) \begin{pmatrix} \Lambda_1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} T_1' \\ T_2' \end{pmatrix} = T_1 \Lambda_1 T_1' = (T_1 \Lambda_1^{1/2})(T_1 \Lambda_1^{1/2})' = BB',$$

where $B = T_1 \Lambda_1^{1/2}$ is a $p \times r$ matrix with rank r .



Exercise 4. Let $A_{n \times n}$ be a positive definite matrix and $B = \begin{bmatrix} A & A \\ A & A \end{bmatrix}$.
Please prove that B is a non-negative matrix.

§ 1.8 Extremal Problem of Eigenvalue

- (1) **(Cauchy–Schwarz Inequation)** If x and y are two p dimensional vectors, then

$$(x'y)^2 \leq (x'x)(y'y) \quad (1.8.1)$$

Equality holds if and only if $y = cx$ with c being some constant.

- (2) **(Extended Cauchy–Schwarz Inequation)** if $B > 0$, then

$$(x'y)^2 \leq (x'Bx)(y'B^{-1}y) \quad (1.8.2)$$

- (3) For any two random variables X and Y , then

$$|E(XY)| \leq E(|XY|) \leq (E(X^2))^{1/2} (E(Y^2))^{1/2}$$

Corollary: $|Cov(X, Y)| \leq Var(X)Var(Y)$

§ 1.8 Extremal Problem of Eigenvalue

- (4) If A is p dimensional symmetric matrix and its eigenvalues are $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$, then

$$\max_{x \neq 0} \mathbf{x}' A \mathbf{x} / \mathbf{x}' \mathbf{x} = \lambda_1 \quad (1.8.3)$$

$$\min_{x \neq 0} \mathbf{x}' A \mathbf{x} / \mathbf{x}' \mathbf{x} = \lambda_p \quad (1.8.4)$$

The equality holds when $x = t_1$ and $x = t_p$, respectively.

- (5) If A is p dimensional symmetric matrix, B is p dimensional positive definite matrix and $\mu_1 \geq \mu_2 \geq \dots \geq \mu_p$ are the eigenvalues of $B^{-1}A$ and the corresponding eigenvectors $t_1, t_2, \dots, t_p$ satisfying $t_i' B t_j = 0, 1 \leq i \neq j \leq p$, then

$$\max_{x \neq 0} \mathbf{x}' A \mathbf{x} / \mathbf{x}' B \mathbf{x} = \mu_1 \quad (1.8.5)$$

$$\min_{x \neq 0} \mathbf{x}' A \mathbf{x} / \mathbf{x}' B \mathbf{x} = \mu_p \quad (1.8.6)$$

The equality holds when $x = t_1$ and $x = t_p$, respectively.

§ 1.8 Extremal Problem of Eigenvalue

- ➡ (2) Proof: Let $y = B^{1/2}x$, $A_1 = B^{-1/2}AB^{-1/2}$, then A_1 and $B^{-1}A$ have the same eigenvalues from Example 1.6.2, denoted as $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_p$. Therefore,

$$\max_{x \neq 0} \frac{x'Ax}{x'Bx} = \max_{x \neq 0} \frac{y'A_1y}{y'y} = \mu_1 \text{ and } \min_{x \neq 0} \frac{x'Ax}{x'Bx} = \min_{y \neq 0} \frac{y'A_1y}{y'y} = \mu_p.$$

Let $u_i = B^{1/2}t_i$, $i = 1, 2, \dots, p$, then

$$A_1 u_i = \left(B^{-\frac{1}{2}} A B^{-\frac{1}{2}} \right) \left(B^{\frac{1}{2}} t_i \right) = B^{\frac{1}{2}} (B^{-1} A t_i) = B^{\frac{1}{2}} (\mu_i t_i) = \mu_i u_i,$$

$$u_i' u_j = \left(B^{\frac{1}{2}} t_i \right)' \left(B^{\frac{1}{2}} t_j \right) = t_i' B t_j = 0, 1 \leq i \neq j \leq p.$$

Thus, $u_1, u_2, \dots, u_p$ is a group of orthogonal eigenvectors of A_1 corresponding to

$\mu_1, \mu_2, \dots, \mu_p$. From (1.8.3), $\frac{x'Ax}{x'Bx}$ ($= \frac{y'A_1y}{y'y}$) has the maximum μ_1

when $x = t_1, y = u_1$ and the minimum μ_p when $x = t_p, y = u_p$.

§ 1.9 Application

- **Exercise 5.** Let $\Sigma > 0$ with order p , for any $p \times p$ symmetric matrix B , please prove that

$$\lambda_{\min}(\Sigma^{1/2} B \Sigma^{1/2}) \geq \lambda_{\min}(B) \lambda_{\min}(\Sigma)$$



Assignments:

See the file of assignment in BB

Deadline: 10:20, March 09, 2026